

Process Model: As-Is and To-Be Analysis

HealthFirst Care Improvement Initiative

This document analyzes HealthFirst Care's current (As-Is) workflows for appointment scheduling, patient check-in, and interdepartmental communication, and proposes optimized (To-Be) workflows to address the inefficiencies identified. It draws on the Stakeholder Profiles document, the appointment, feedback, and resource data files, and the objectives and requirements already established in the Business Requirements Document (BRD) and Requirements Traceability Matrix (RTM).

1. Current Process Analysis — Key Challenges

1.1 Delays in Appointment Scheduling

Appointment scheduling is handled manually by phone or in person, with administrative staff (Maria Carter) checking doctor availability by hand — a process with no automated conflict detection. This matches what the data shows: in `appointment_data.csv` (n=20), only 8 appointments (40%) were marked Completed; the remaining 12 (60%) were split evenly across No Show, Rescheduled, and Cancelled (4 each, 20% apiece). Each department also runs on a single fixed time slot in the sample data, consistent with the double-booking complaints raised by administrative and clinical stakeholders.

1.2 Inefficient Patient Check-In

Check-in relies on paper intake forms that staff re-key into the records system by hand, then patients wait in an unstructured queue for manual verification. `feedback_data.csv` (n=15) shows an overall average satisfaction of 7.0/10, but General Medicine (5.33) and Paediatrics (6.0) score well below that average, with written comments citing "long wait times," "delayed response," and a "room cleanliness issue" — consistent with a check-in process that has no digital queue or automated verification step.

1.3 Interdepartmental Communication Issues

When administrative staff identify a resource need (e.g., equipment, a nurse, a room), the request is currently made by phone or email directly to IT, with no shared tracking system — so requests can sit unresolved with no audit trail. `resource_data.csv` (n=10) shows this concretely: two Nurse-type records are logged "Unavailable" with 0 usage hours, and two Room-type records are logged "Under Maintenance," limiting throughput exactly when Doctors (Dr. Khan, Dr. Lee) and Nurses (Santa Murmu, Jessica Gomes) report needing that capacity most.

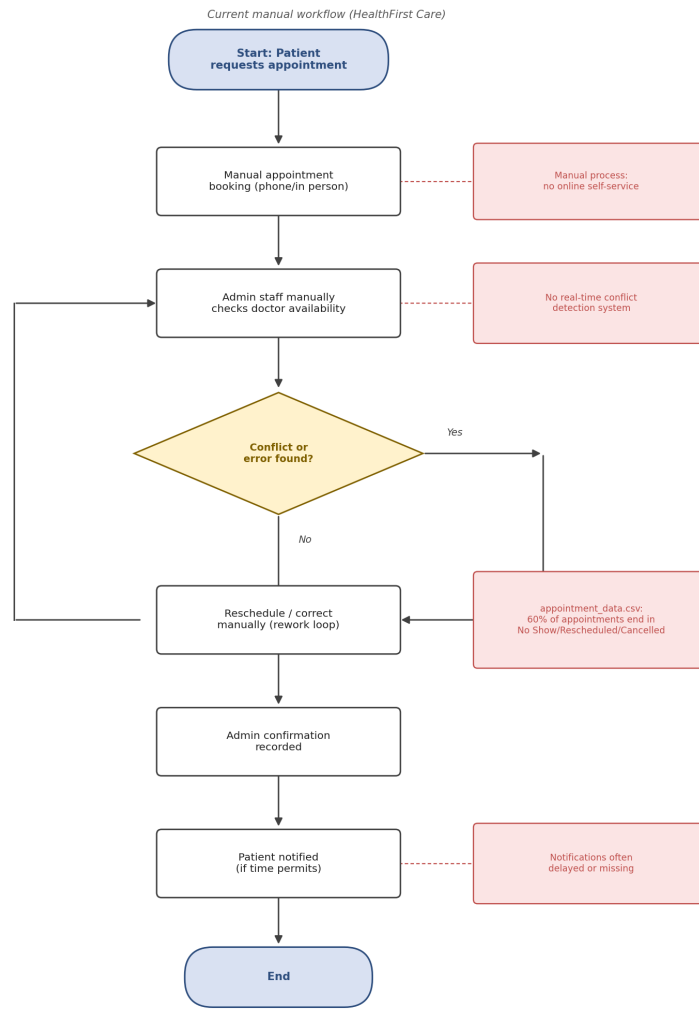
Simulated stakeholder feedback used to identify these challenges:

- Maria Carter (Appointment Scheduler): frequent double bookings due to system glitches; limited visibility into doctors' real-time availability.
- Lak Ayer (Patient): found the online scheduling system confusing; waited nearly an hour for a cardiology consultation.
- Rajesh Singh (IT Manager): scheduling and record-management systems are not integrated, creating data silos and slow resource allocation.
- Jessica Gomes (ER Nurse): delays receiving diagnostic results from radiology; miscommunication during ICU patient transfers causing duplicated effort.

2. As-Is Process Models

The three current-state workflows below are attached in full resolution as As-Is_Process_Models.pdf / .jpg. Bottlenecks are highlighted in red callouts on each diagram.

As-Is Process Model: Appointment Scheduling



 Bottleneck / Pain point

Figure 1 — As-Is: Appointment Scheduling

As-Is Process Model: Patient Check-In

Current manual workflow (HealthFirst Care)

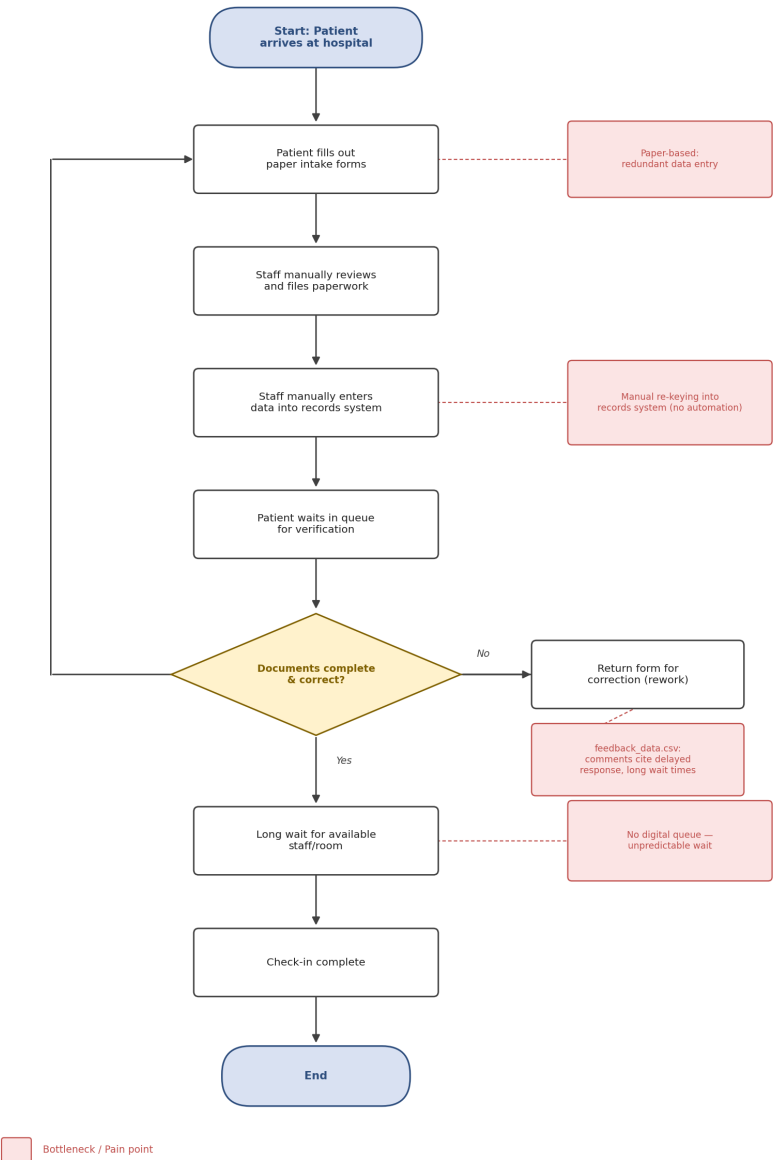
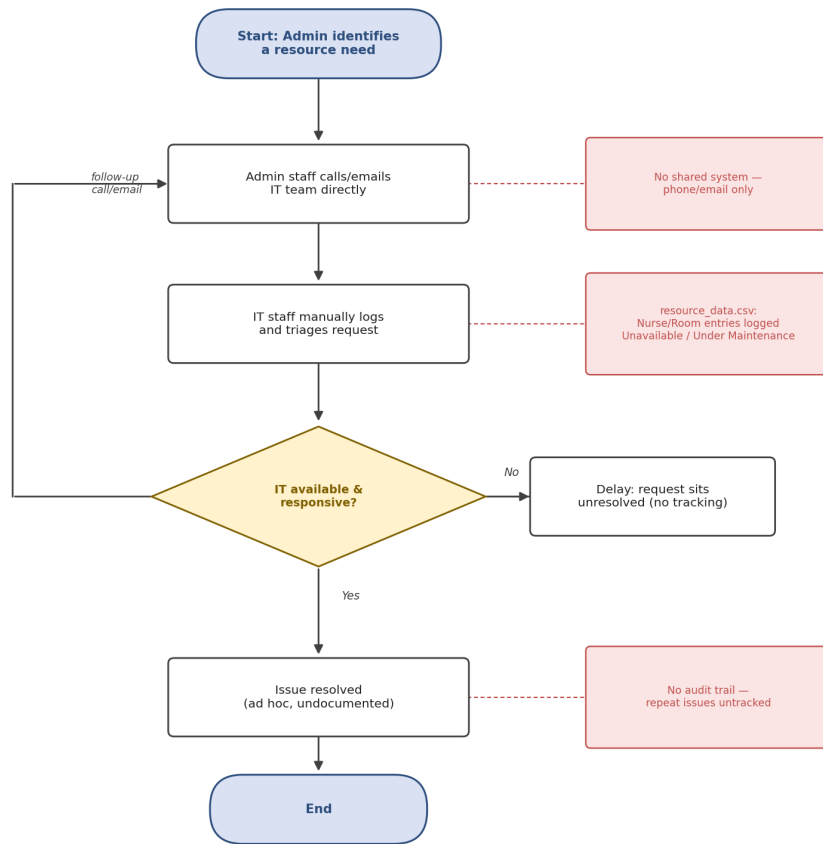


Figure 2 — As-Is: Patient Check-In

As-Is Process Model: Interdepartmental Communication

Admin → IT resource-request workflow (HealthFirst Care)



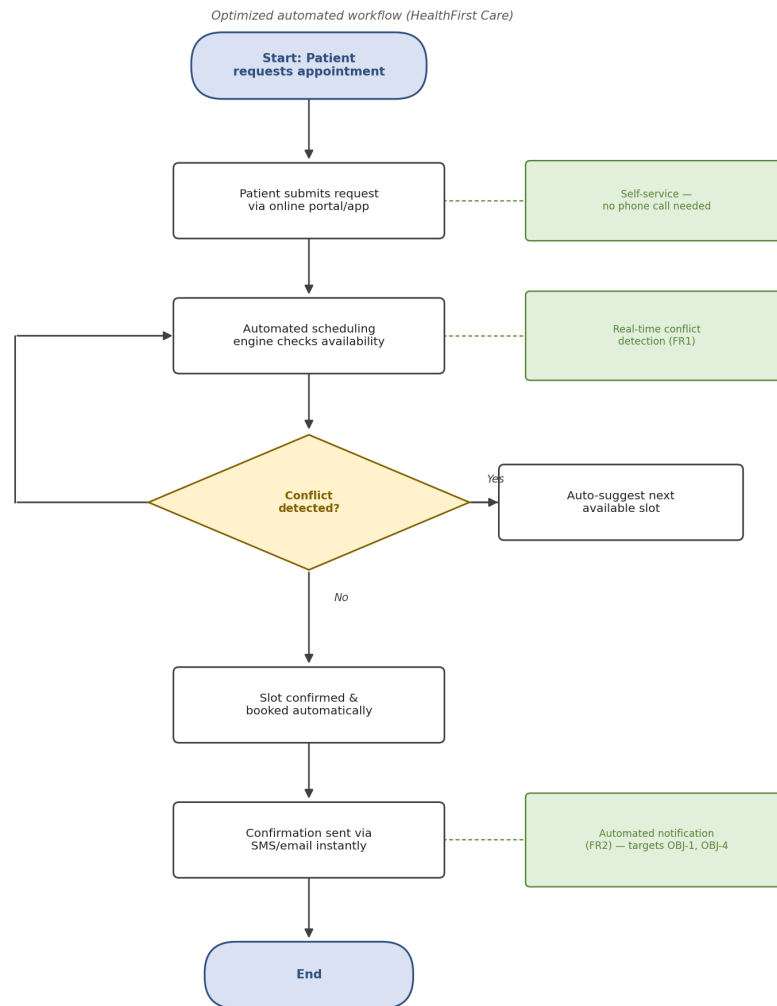
Bottleneck / Pain point

Figure 3 — As-Is: Interdepartmental Communication

3. To-Be Process Models

The optimized workflows below are attached in full resolution as To-Be_Process_Models.pdf / .jpg. Improvements are highlighted in green callouts, with references to the BRD requirement(s) and business objective(s) each one supports.

To-Be Process Model: Appointment Scheduling



 Improvement

Figure 4 — To-Be: Automated Appointment Scheduling

To-Be Process Model: Patient Check-In

Optimized self-service workflow (HealthFirst Care)

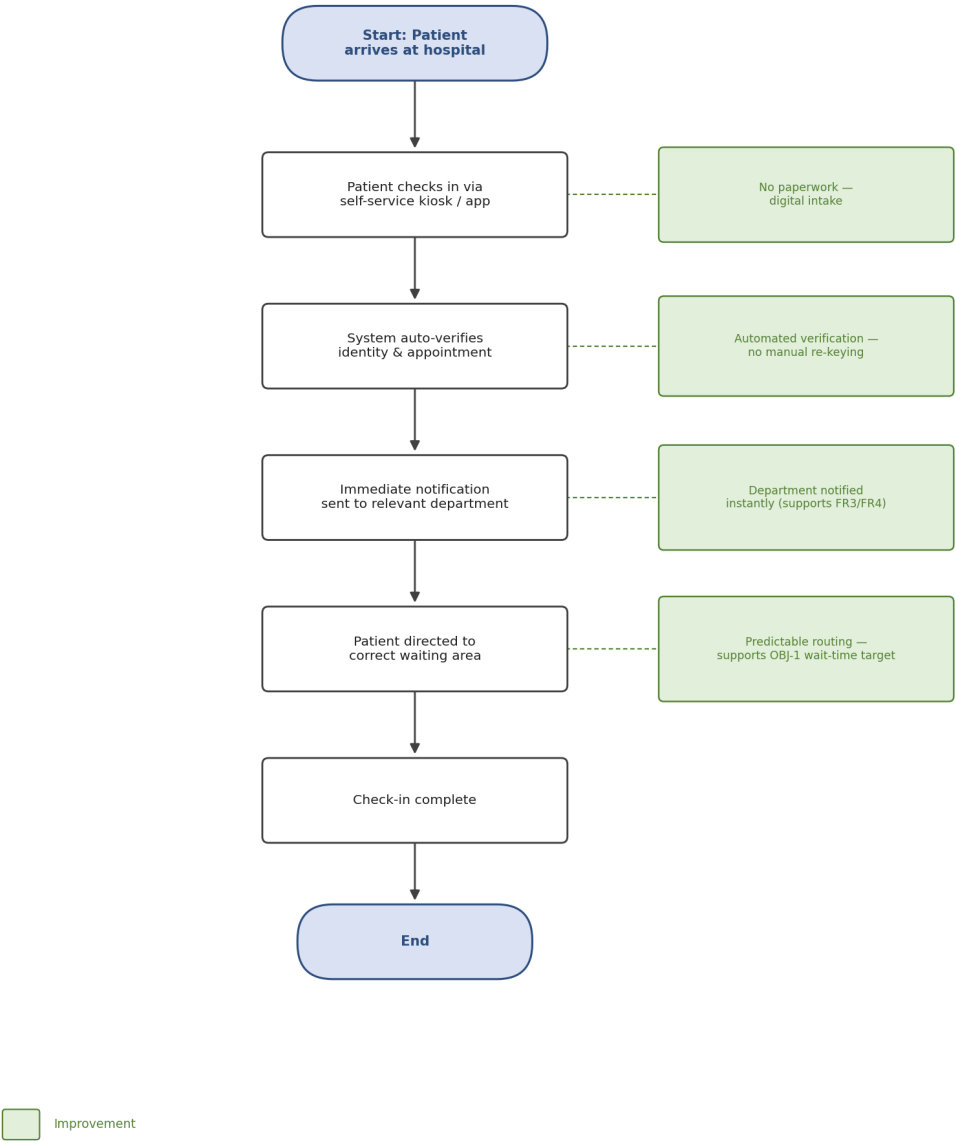


Figure 5 — To-Be: Streamlined Patient Check-In

To-Be Process Model: Interdepartmental Communication

Automated task-management workflow (HealthFirst Care)

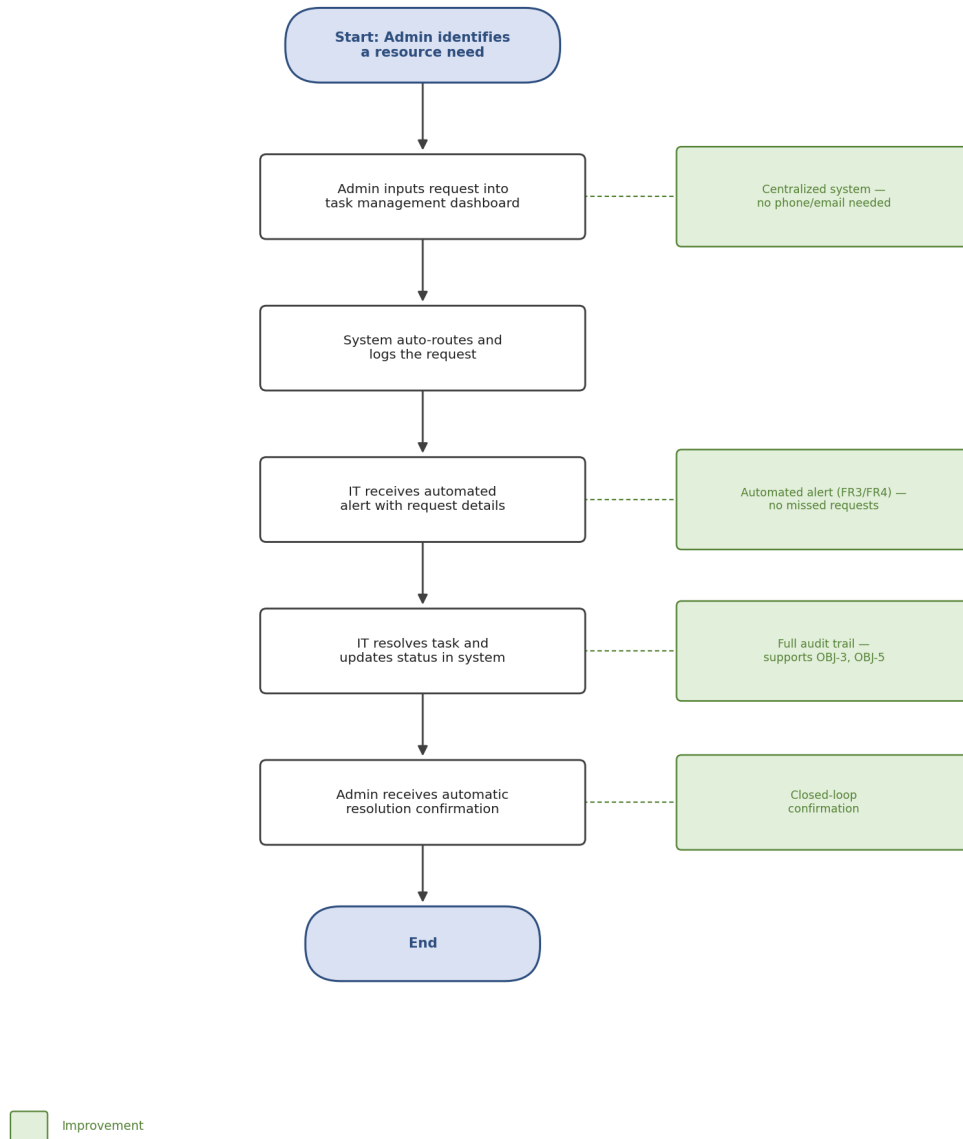


Figure 6 — To-Be: Automated Interdepartmental Communication

Rationale for Proposed Solutions & Expected Impact

Workflow	Proposed Solution	Rationale	Expected Impact
Appointment Scheduling	Automated scheduling engine with real-time conflict detection and instant SMS/email confirmation (FR1, FR2)	Removes the manual availability check that causes double bookings; directly targets the 60% non-completion rate observed in appointment_data.csv	Supports OBJ-1 (20% wait-time reduction) and OBJ-2 (fewer double bookings)
Patient Check-In	Self-service kiosk/app check-in with automated verification and instant department notification	Removes paper intake and manual re-keying, the two steps most linked to the wait-time complaints in feedback_data.csv	Supports OBJ-1 and OBJ-4 (patient satisfaction), especially in General Medicine and Paediatrics

Workflow	Proposed Solution	Rationale	Expected Impact
Interdepartmental Communication	Centralized task-management dashboard with automated IT alerts and closed-loop confirmation (FR3, FR4)	Replaces untracked phone/email requests with a system of record, addressing the Unavailable/Under Maintenance patterns in resource_data.csv	Supports OBJ-3 (communication/handoff efficiency) and OBJ-5 (eliminating data silos)
<i>Note on evidence: the data files reviewed are small samples (10–20 rows each) provided for this exercise, so the figures above should be read as directionally consistent with the stakeholder-reported problems rather than statistically conclusive on their own.</i>			

4. Attachments
- As-Is_Process_Models.pdf / As-Is_Process_Models.jpg — current-state flowcharts for all three workflows (also embedded above). - To-Be_Process_Models.pdf / To-Be_Process_Models.jpg — optimized-state flowcharts for all three workflows (also embedded above).